STAT6011

COMPUTATIONAL STATISTICS AND BAYESIAN LEARNING

Lecture 2

Bootstrap

Bootstrap

- Suppose that we have a sample of n observations, $y_1, \ldots, y_n$.
- Denote the probability density function (PDF) f, and the cumulative distribution function (CDF) F.
- We use the sample to make inferences about the population parameter θ.
- How to compute the variance and confidence interval for θ?

Bootstrap

- Basic idea: our sample gives us all the information we have about the whole population.
  1. Calculate the statistic of interest (call it $\hat{\theta}$) from the sample.
  2. Find the CDF $\hat{F}$, draw a "bootstrap sample" from $\hat{F}$ and calculate the statistic of interest based on bootstrap samples.
     - ✓ Draw a sample of size n from the original dataset **with replacement**
     - ✓ $Y_1^*, \ldots, Y_n^* \sim \hat{F}$
     - ✓ $\hat{\theta}^* = \hat{\theta}(Y_1^*, \ldots, Y_n^*)$

3. Repeat step 2 independently for a large number B of times obtaining bootstrap replications $\hat{\theta}^{*1}, \ldots, \hat{\theta}^{*B}$.

- Use bootstrap replications to:
  - ✓ estimate the variance and standard error of $\hat{\theta}$
  - ✓ estimate the bias
  - ✓ obtain the confidence interval

Percentile Method for Confidence Intervals

- Denote the CDF of the bootstrap distribution of $\hat{\theta}^*$ as

$$F_{\hat{\theta}^*}(t) = P(\hat{\theta}^* \leq t).$$

- If the bootstrap distribution is obtained by simulations, then

$$\widehat{F}_{\hat{\theta}^*}(t) \approx \frac{\sum_{b=1}^{B} I(\hat{\theta}^{*b} \leq t)}{B}.$$

- Define the confidence interval as the interval between appropriate quantiles/percentiles.

Parametric Bootstrap

1. Estimate the parametric MLE of the unknown F, i.e., obtain the MLEs of the parameters $\hat{\theta}$, and find $\hat{F}_{\hat{\theta}}$.
2. Draw a "bootstrap sample" from $\hat{F}_{\hat{\theta}}$ and calculate statistic of interest based on bootstrap samples.
   - ✓ Simulate data from the parametric model using MLEs as parameters.
   - ✓ $Y_1^*, \ldots, Y_n^* \sim \hat{F}_{\hat{\theta}}$.
   - ✓ $\hat{\theta}^* = \hat{\theta}(Y_1^*, \ldots, Y_n^*)$.
3. Repeat step 2 independently a large number B of times, and obtain bootstrap replications $\hat{\theta}^{*1}, \ldots, \hat{\theta}^{*B}$.

Bootstrap in Regression

- Consider the observed data $\{(x_1, x_2, y)_i; i = 1, \ldots, n\}$. The linear model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon,$$

where $E(\epsilon) = 0$, var$(\epsilon) = \sigma^2$, and the ϵ_i's are independent.

- Parametric bootstrap:
  - ✓ Obtain the MLEs $\hat{\beta}_0$, $\hat{\beta}_1$, $\hat{\beta}_2$, and $\hat{\sigma}^2$.
  - ✓ Sample $\epsilon_i^* \sim$ iid N$(0, \hat{\sigma}^2)$.
  - ✓ Take $y_i^* = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \epsilon_i^*$.
  - ✓ Obtain $\hat{\beta}_0^*$, $\hat{\beta}_1^*$, $\hat{\beta}_2^*$ and $\hat{\sigma}^{*2}$, and repeat many times.

- Nonparametric bootstrap version I:
  - ✓ Sample $(x_1, x_2, y)_i^*$ by drawing with replacement from $(x_1, x_2, y)_i$.
  - ✓ Obtain $\hat{\beta}_0^*$, $\hat{\beta}_1^*$, and $\hat{\beta}_2^*$ and repeat many times.
- Nonparametric bootstrap version II:
  - ✓ Obtain $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$.
  - ✓ Calculate $\hat{\epsilon}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2}$.
  - ✓ Sample ϵ_i^* by drawing with replacement from $\{\hat{\epsilon}_i\}$.
  - ✓ Obtain $\hat{\beta}_0^*$, $\hat{\beta}_1^*$, and $\hat{\beta}_2^*$ and repeat many times.
- The bootstrap procedure of resampling with replacement mimics the process of taking samples from the population, so it can provide us with new information about the variation in these samples, and so about the sampling error.

Permutation Test

- **Data**: $(\mathbf{x}, y)_i$ for $i = 1, \ldots, n$
- **Question**: Are the $\mathbf{x}$'s and y's associated?
- **Test Statistic**: $T(\mathbf{x}, y)$ (a large value of T implies high association)
- **Permutation distribution**: Obtain the distribution of $T(\mathbf{x}^*, y^*)$, where $(\mathbf{x}^*, y^*)$ is a permuted version of $(\mathbf{x}, y)$, producing the null distribution of the test statistics (i.e., H_0 : no association).

- Permutation destroys the association between $\mathbf{x}$'s and y's.
- By generating a large number of permutation samples, we compute the test statistics for each sample,

$$\text{p-value} = P[T(\mathbf{x}^*, y^*) > T(\mathbf{x}, y)]$$

- The key step in the general procedure for permutation tests is to form permutation resamples in a way that is consistent with the study design and with the null hypothesis.

Efron's Bootstrap

- Suppose that we have a sample of size n, with observed data $Z_1, \ldots, Z_n$, where the Z_i's are independent, identically distributed realizations from some distribution $F \in \mathcal{F}$.
- The parameter of interest is a functional $T(F)$ where T is a mapping from $\mathcal{F}$ to $\mathcal{R}$, e.g., the mean of the distribution, $\mu = \int z dF$.
- Let F_n be

$$F_n = \sum_{i=1}^{n} w_i \delta_{Z_i},$$

where δ_{Z_i} is a degenerate probability measure at Z_i, and w_i satisfies $\sum_{i=1}^{n} w_i = 1$.

- The bootstrap distribution of $T(F)$ is based on considering all possible bootstrap replications $T(F_n^{(r)})$ where $w_i^{(r)}$ in $F_n^{(r)}$ corresponds to the proportion of times Z_i appears in the rth bootstrap replicate, with $w_i^{(r)}$ taking on values in $\{0, 1/n, \ldots, n/n\}$.
- The bootstrap mean for the rth replicate is calculated as

$$\hat{Z}^{(r)} = \sum_{i=1}^{n} w_i^{(r)} Z_i.$$

- The bagging (bootstrap aggregating) estimate of the mean is the average over all bootstrap replicates, $\frac{1}{R}\sum_{r=1}^{R} \hat{Z}^{(r)}$, where R is the number of bootstrap replicates.

Bayesian Bootstrap

- Instead of drawing weights w_i from the discrete set $\{0, 1/n, \ldots, n/n\}$, the Bayesian approach treats the vector of weights w in F_n as unknown parameters and derives a posterior distribution for w, and hence $T(F)$.
- Rubin (1981) used a non-informative prior, $\prod_{i=1}^{n} w_i^{-1}$, which, when combined with the multinomial likelihood for Z, leads to a Dirichlet$(1, \ldots, 1)$ distribution for the posterior distribution of w.
- The posterior distribution of $T(F)$ is estimated by Monte Carlo method: generate $w^{(b)}$ from a Dirichlet$(1, \ldots, 1)$ distribution and then calculate $T(F_n^{(b)})$ for each sample $w^{(b)}$.

- The average of $T(F_n^{(b)})$ over the samples corresponds to the Monte Carlo estimate of the posterior mean of $T(F)$ and can be viewed as a Bayesian analog of bagging.
- Rubin's Bayesian bootstrap can be viewed as a limiting case of the nonparametric distribution for F using a Dirichlet process prior (Gasparini, 1995).
- If the prior for F is a Dirichlet process with parameter α, $DP(\alpha, F_0)$, then the posterior distribution for F is again a Dirichlet process $DP(\alpha + n, (\alpha + n)^{-1}(\alpha F_0 + \sum_{i=1}^{n} \delta_{Z_i}))$ (Ferguson, 1973).
- In the limit as α goes to 0, the posterior distribution is a $DP(n, n^{-1} \sum_{i=1}^{n} \delta_{Z_i})$, with mean equal to the empirical cumulative distribution function.

- It is this limiting noninformative case that is equivalent to Rubin's Bayesian bootstrap (Gasparini, 1995).

Jackknife

- Another resampling-based method of trying to assess bias, standard error, etc. of a sample-based estimate $\hat{\theta}$ of a population characteristic θ.
- Usually less computationally intensive than the bootstrap
- Let n denote sample size of real dataset
- There are n jackknife samples $\mathbf{X}_{(-i)}$, each obtained by leaving out one observation i from the original dataset
  $\mathbf{X}_{(-i)} = \{x_1, x_2, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n\}$
- Each *jackknife replication* $\hat{\theta}_{(i)}$ is calculated from the jackknife sample $\mathbf{X}_{(-i)}$ in the same way that $\hat{\theta}$ was calculated from the whole dataset
- The ith jackknife *pseudo-value* $\tilde{\theta}_i$ is calculated as $n\hat{\theta} - (n-1)\hat{\theta}_{(i)}$

Jackknife estimate of standard error of $\hat{\theta}$

- Let $\hat{\theta}_{(.)}$ denote the mean of the jackknife replication $\hat{\theta}_{(i)}$. Then

$$\widehat{\mathrm{SE}}_{\mathrm{jack}} = \sqrt{\frac{n-1}{n}\sum_{i=1}^{n}(\hat{\theta}_{(i)} - \hat{\theta}_{(.)})^2} = \sqrt{\frac{\sum_{i=1}^{n}(\tilde{\theta}_i - \tilde{\theta}_.)^2}{n(n-1)}}$$

- Multiplicative factor $\frac{n-1}{n}$ is chosen to make this work out exactly right for the case of estimating a population mean using $\hat{\theta} = \bar{x}$
- $\widehat{\mathrm{Bias}}_{\mathrm{jack}} = (n-1)(\hat{\theta}_{(.)} - \hat{\theta})$
- Multiplicative factor $(n-1)$ is chosen to make this work out exactly right for the case of estimating a population variance using $\hat{\sigma}^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}$.

- The (first order) jackknife estimator of θ is

$$\hat{\theta}^{\text{jack}} = n\hat{\theta} - (n-1)\hat{\theta}_{(.)} = \frac{\sum_{i=1}^{n} \tilde{\theta}_i}{n}$$

- Why can we expect $\hat{\theta}^{\text{jack}}$ to be less biased than $\hat{\theta}$
  - ✓ Most estimators are biased
  - ✓ Bias can be approximated as Taylor series expansion of the estimator

$$E(\hat{\theta} - \theta) = \frac{a_1}{n} + \frac{a_2}{n^2} + \frac{a_3}{n^3} + \dots$$

  - ✓ It can be shown that

$$E(\hat{\theta}^{\text{jack}} - \theta) = -\frac{a_2}{n^2} - \frac{a_2 + a_3}{n^3} - \dots$$

    which eliminates term of order $\frac{1}{n}$ and thus is smaller than the bias of $\hat{\theta}$.

Relationship between jackknife and bootstrap

- Unless n is large, jackknife is less computationally intensive
- But if jackknife uses fewer samples than bootstrap, then jackknife is using less information
- Jackknife may be considered an approximation to the bootstrap
- Jackknife does not work well if the quantity being estimated is not “smooth”, e.g., median can change sharply with deletion of a single observation.